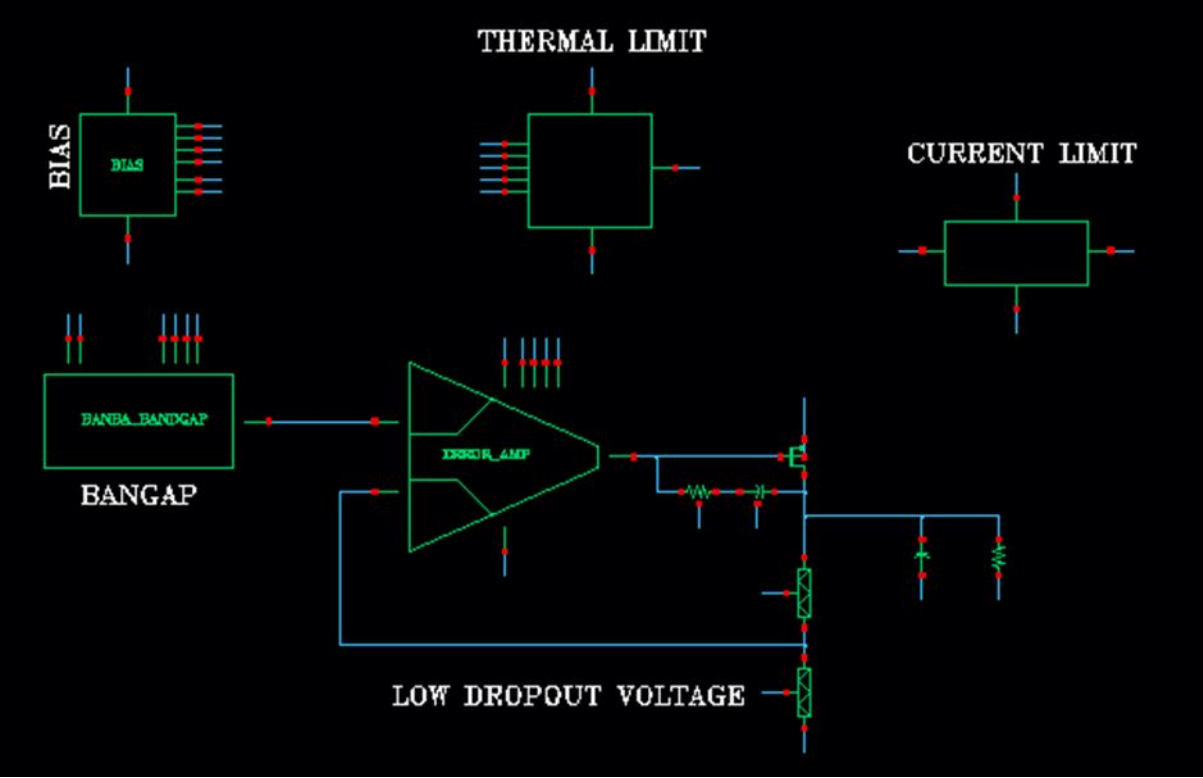


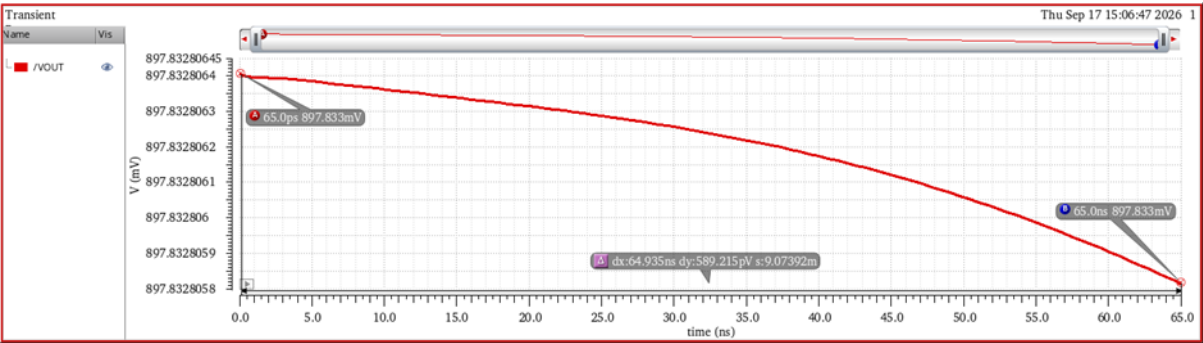
Họ tên: Lê Minh Hiền

Trường: Đại học Công Nghiệp Hà Nội

Low Dropout Voltage 45nm



Iload max = 5mA

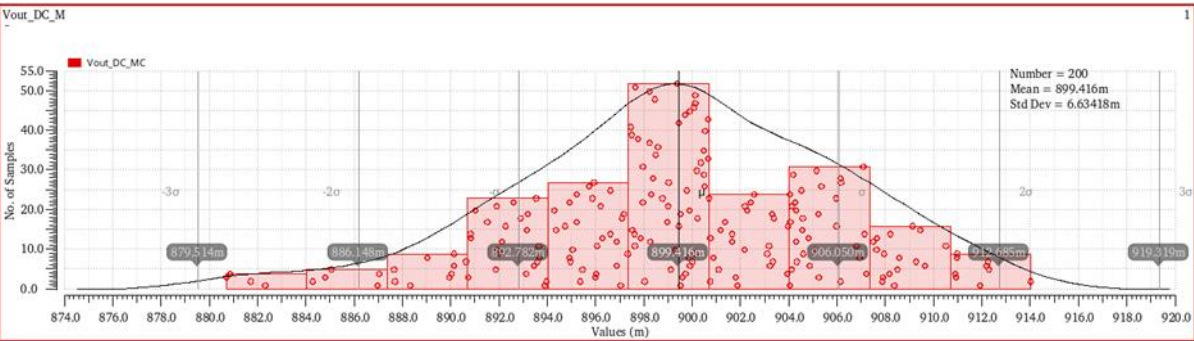


Test	Output	Spec	Weight	Pass/Fail	Min	Max
Filter	Filter	Filter	Filter	Filter	Filter	Filter
vout	Vout_DC				889.2m	902.8m
vout	Vout_Max	< 0.909		pass	889.2m	902.8m
vout	Vout_Min	> 0.891		near	889.2m	902.8m
vout	Delta_Vout				144.3p	2.091n
vout	/Vbandgap				297.6m	301.5m

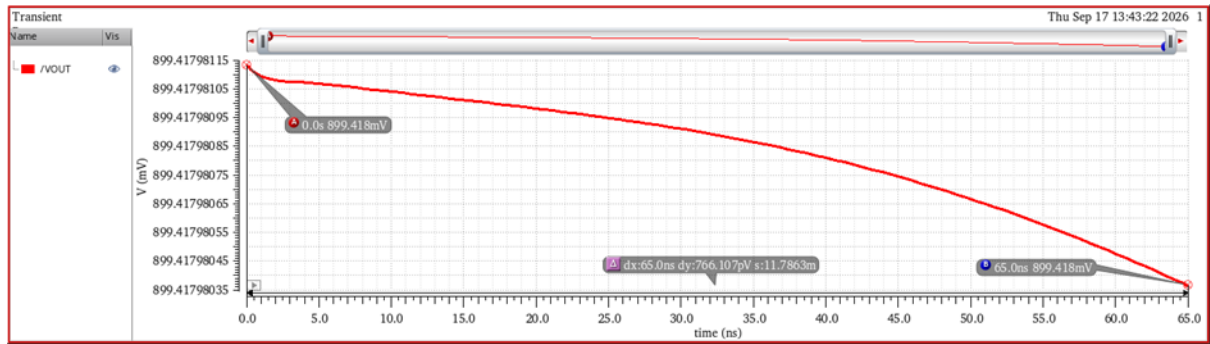
Test	Name	Yield	Min	Target	Max	Mean	Std Dev	Cpk	Errors
Yield Estimate: 100 %(200 passed/200 pts) Confidence Level: <not set> Filter: <not set>									
-	vout								
-	Vout_DC(summary)	100	880.7m		914m	899.4m	6.634m		0
	Vout_DC_MC	100	880.7m	info	914m	899.4m	6.634m		0
-	Delta_Vout(summary)	100	522.1p		837p	670.7p	57.94p		0
	Delta_Vout_MC	100	522.1p	info	837p	670.7p	57.94p		0
-	/Vbandgap(summary)	100	293.9m		305.5m	300.4m	2.226m		0
	/Vbandgap_MC	100	293.9m	info	305.5m	300.4m	2.226m		0

	Max	Vout_DC MC R^2 = 0.90503	Delta_Vout MC R^2 = 0.90206	/Vbandgap MC R^2 = 0.91433
Top	100%	100%	100%	100%
Process	96%	19%	96%	17%
/I13/PM1	24%	22%	0%	24%
/I13/I20/PM5_2	12%	10%	0%	12%
/I13/PM2	11%	11%	0%	10%
/I13/I20/NM2_1	10%	10%	0%	8%
/I13/I20/NM2_2	9%	8%	0%	9%
/I13/I20/PM5_1	8%	8%	0%	7%
/I13/I20/PM1_1	5%	5%	0%	5%
/I13/I20/PM1_2	5%	4%	0%	5%

	Average(Mag)	Vout_DC MC R^2 = 0.90503 (Top 100.0%)	Delta_Vout MC R^2 = 0.90206 (Top 100.0%)	/Vbandgap MC R^2 = 0.91433 (Top 100.0%)
I13.PM1xp1	13%	18%	0%	20%
statis_bjt_pnp	7%	11%	0%	11%
I13.I20.NM2_1rn2	5%	8%	0%	8%
I13.I20.PM5_1xp1	5%	8%	0%	7%
I13.I20.PM5_2xp1	5%	8%	0%	9%
I13.PM2xp1	5%	8%	0%	8%
random_dsw_resnsmptly_dis	22%	7%	53%	6%
I13.I20.NM2_2rn2	4%	6%	0%	7%
I13.I20.PM1_1xp1	3%	5%	0%	5%



Iload min = 800u A:

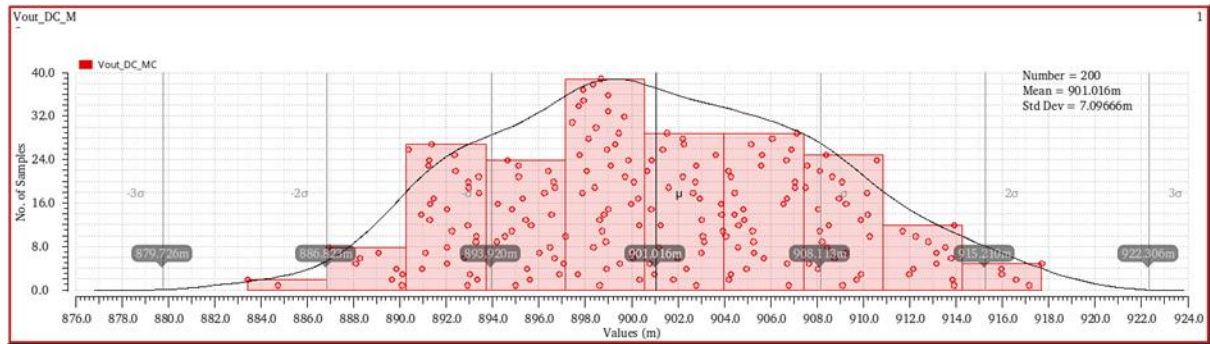


Test	Output	Spec	Weight	Pass/Fail	Min	Max
Filter	Filter	Filter	Filter	Filter	Filter	Filter
vout	Vout_DC				892.9m	904.3m
vout	Vout_Max	< 0.909		pass	892.9m	904.3m
vout	Vout_Min	> 0.891		pass	892.9m	904.3m
vout	Delta_Vout				226.3p	2.509n
vout	/Vbandgap				297.6m	301.5m

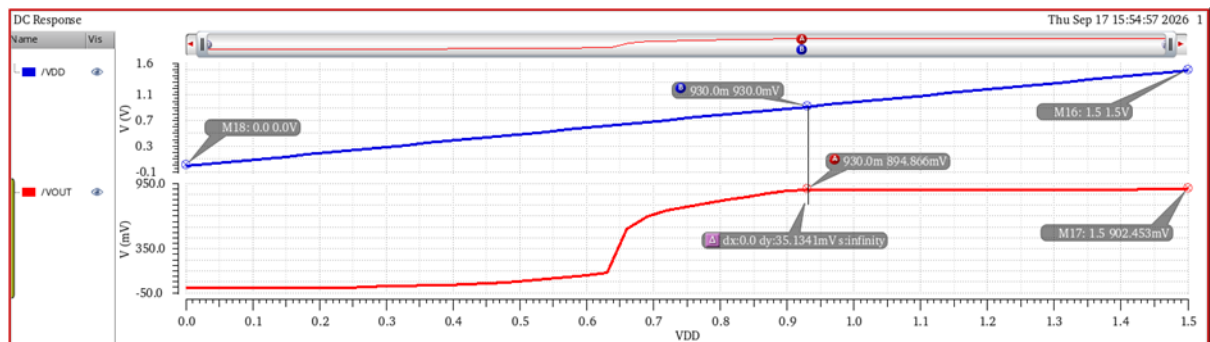
Test	Name	Yield	Min	Target	Max	Mean	Std Dev	Cpk	Errors
Yield Estimate: 100 %(200 passed/200 pts) Confidence Level: <not set> Filter: <not set>									
- vout									
	Vout_DC(summary)	100	883.4m		917.7m	901m	7.097m		0
	Vout_DC_MC	100	883.4m	info	917.7m	901m	7.097m		0
	Delta_Vout(summary)	100	990.2p		1.465n	1.19n	92.47p		0
	Delta_Vout_MC	100	990.2p	info	1.465n	1.19n	92.47p		0
	/Vbandgap(summary)	100	294.8m		306.4m	300.4m	2.308m		0
	/Vbandgap_MC	100	294.8m	info	306.4m	300.4m	2.308m		0

	Max	Vout_DC MC R ² = 0.90290	Delta_Vout MC R ² = 0.90698	/Vbandgap MC R ² = 0.91138
Top	100%	100%	100%	100%
Process	100%	14%	100%	12%
/I13/PM1	28%	28%	0%	27%
/I13/PM2	17%	11%	0%	17%
/I13/I20/NM2_1	10%	9%	0%	10%
/I13/I20/PM5_2	10%	10%	0%	7%
/I13/I20/NM2_2	8%	8%	0%	5%
/I13/I20/PM1_1	8%	7%	0%	8%
/I13/I20/PM5_1	8%	5%	0%	8%
/I13/I20/PM1_2	6%	5%	0%	6%

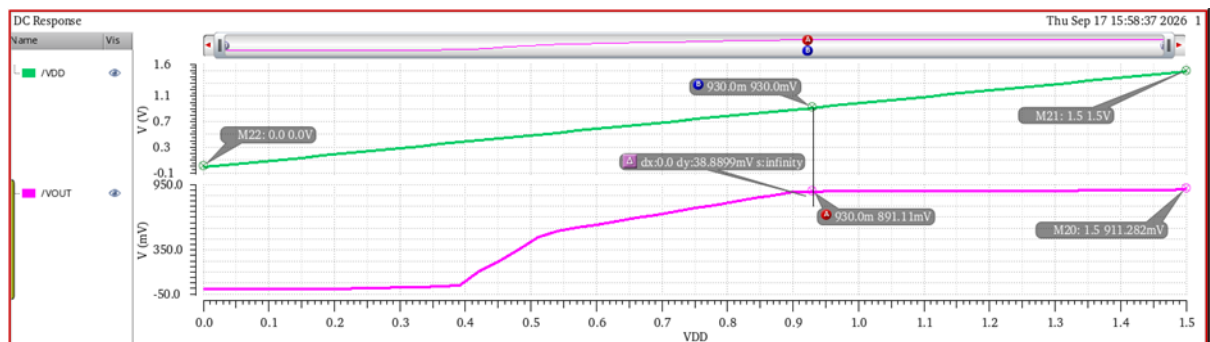
	Average(Mag)	Vout_DC MC R ² = 0.90290 (Top 100.0%)	Delta_Vout MC R ² = 0.90698 (Top 100.0%)	/Vbandgap MC R ² = 0.91138 (Top 100.0%)
I13.PM1xp1	14%	22%	0%	21%
statis_bjt_pnp	7%	11%	0%	10%
I13.I20.PM5_2xp1	6%	10%	0%	7%
I13.PM2xp1	7%	9%	0%	13%
I13.I20.PM1_1xp1	5%	7%	0%	8%
I13.I20.NM2_1xn2	4%	6%	0%	7%
I13.I20.NM2_2xn2	4%	6%	0%	5%
I13.PM1xp2	4%	6%	0%	5%
I13.I20.PM1_2xp1	4%	5%	0%	6%



Drop Voltage:

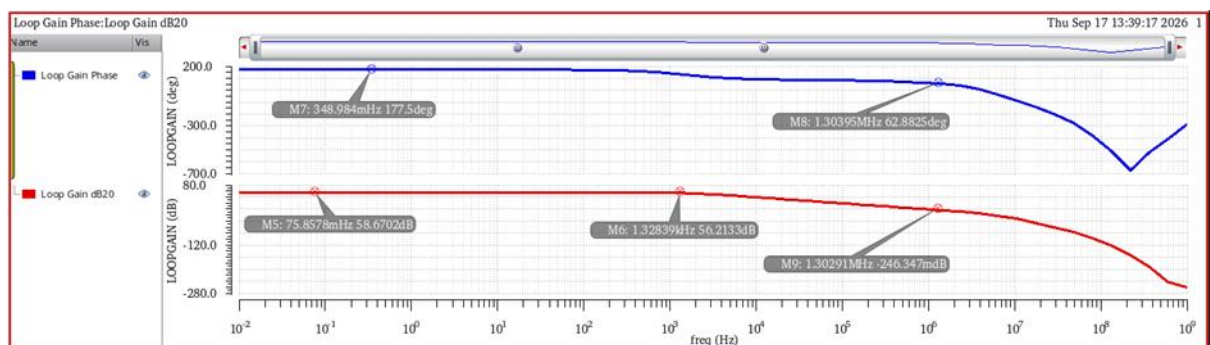


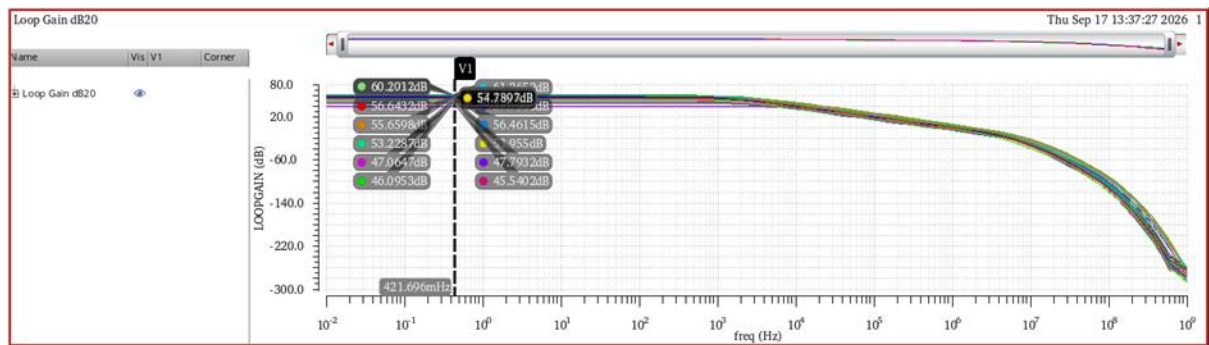
Worst corner:



Loop:

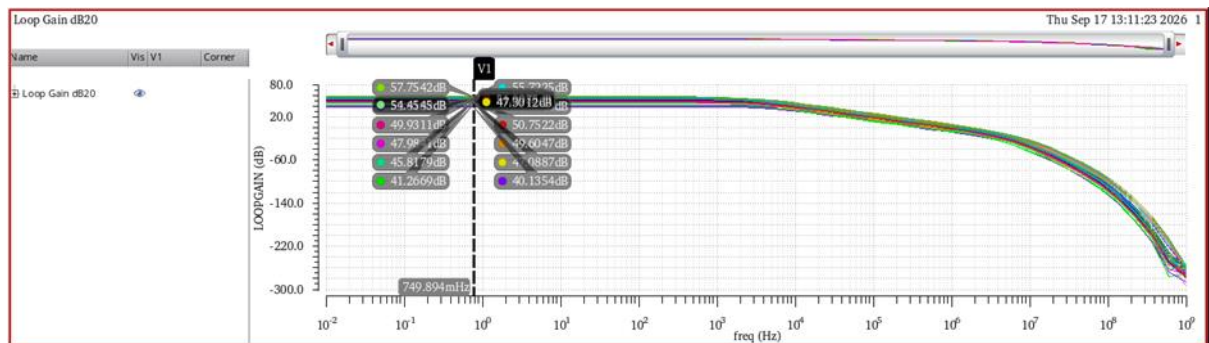
$I_{min} = 800\mu A$





Test	Output	Spec	Weight	Pass/Fail	Min	Max
Filter	Filter	Filter	Filter	Filter	Filter	Filter
stb	Loop Gain dB20					
stb	Gain Loop	> 40		pass	40.29	62.64
stb	Phase Margin	> 45		pass	49.15	79.42

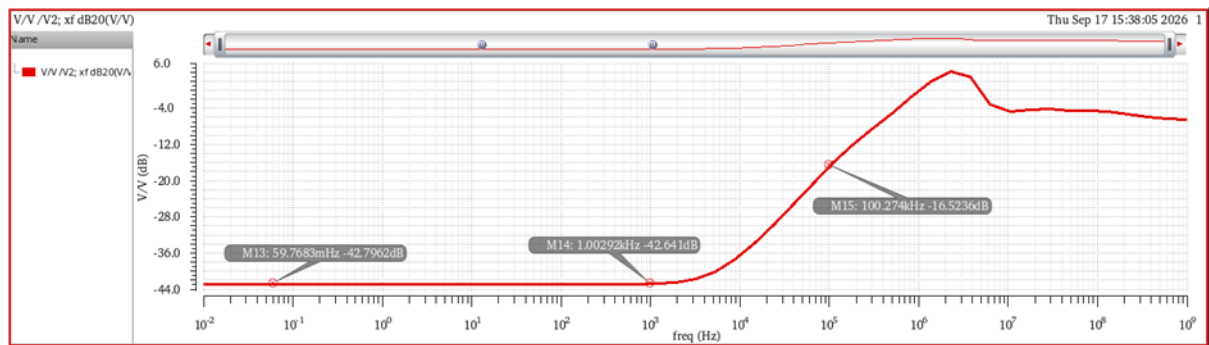
$I_{max} = 5mA$



Test	Output	Spec	Weight	Pass/Fail	Min	Max
Filter	Filter	Filter	Filter	Filter	Filter	Filter
stb	Loop Gain dB20					
stb	Gain Loop	> 40		pass	40.14	59.22
stb	Phase Margin	> 45		pass	50.14	79.98

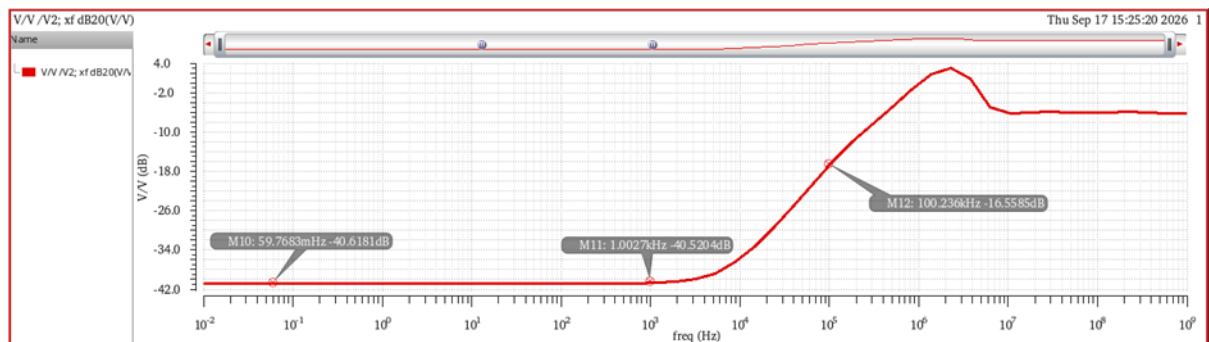
PSRR:

$I_{min} = 800uA$



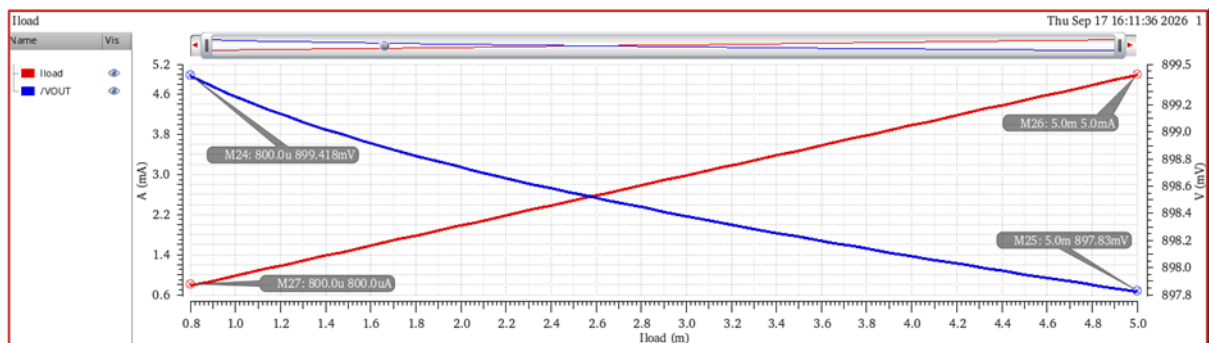
Test	Output	Spec	Weight	Pass/Fail	Min	Max
Filter	Filter	Filter	Filter	Filter	Filter	Filter
psrr	Psrr_DC				-46.93	-27.99
psrr	Psrr_1k				-46.8	-27.98
psrr	Psrr_100k				-21.73	-12.23

$I_{max}=5mA$

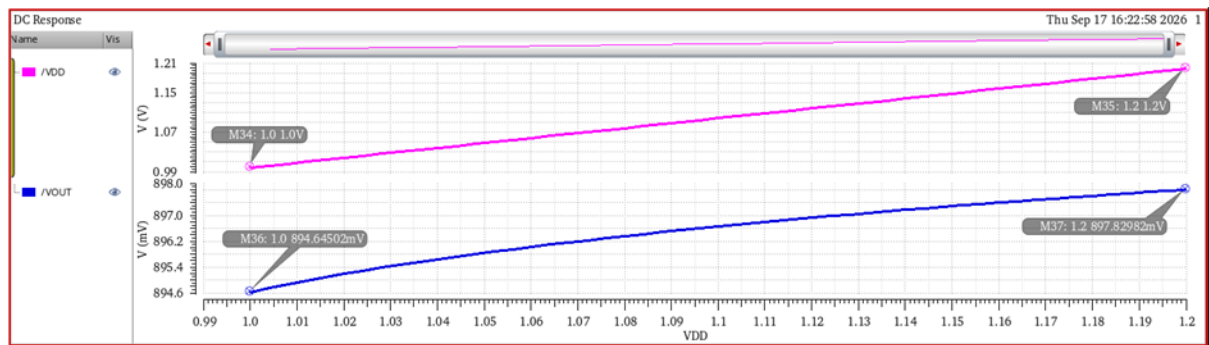


Test	Output	Spec	Weight	Pass/Fail	Min	Max
Filter	Filter	Filter	Filter	Filter	Filter	Filter
psrr	Psrr_DC				-45.76	-27.4
psrr	Psrr_1k				-45.66	-27.39
psrr	Psrr_100k				-21.66	-10.93

Load Regulation:



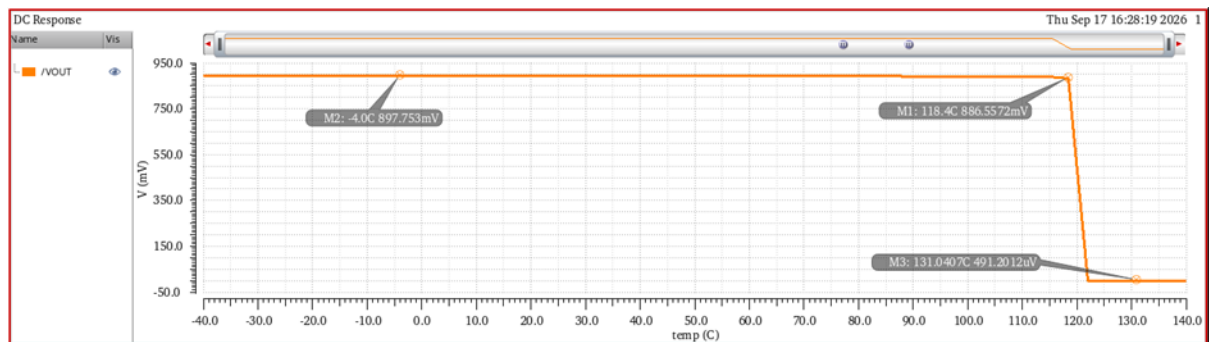
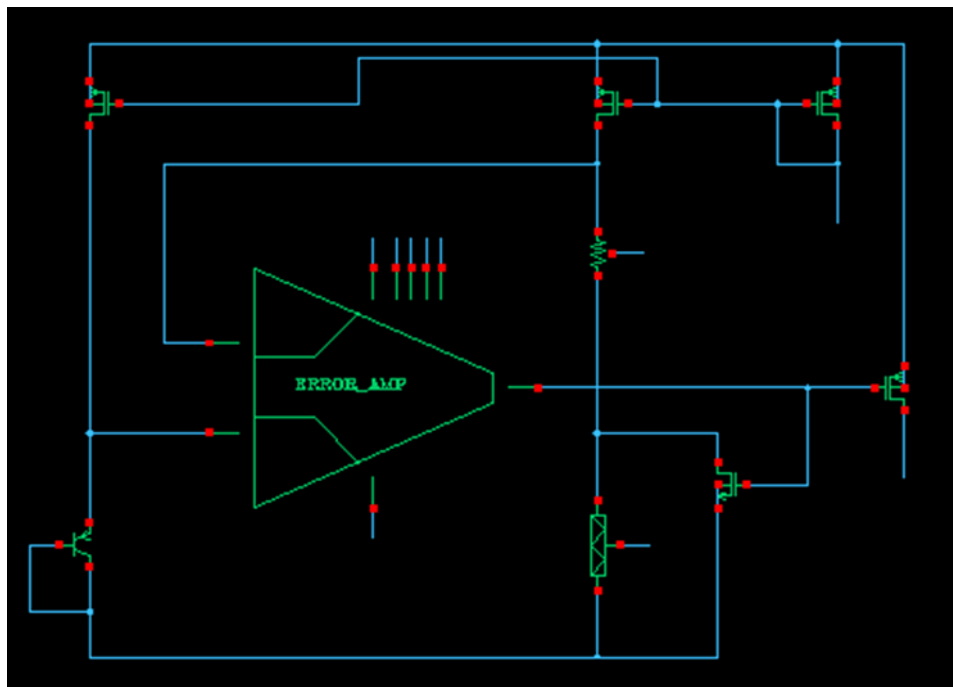
Line Regulation:

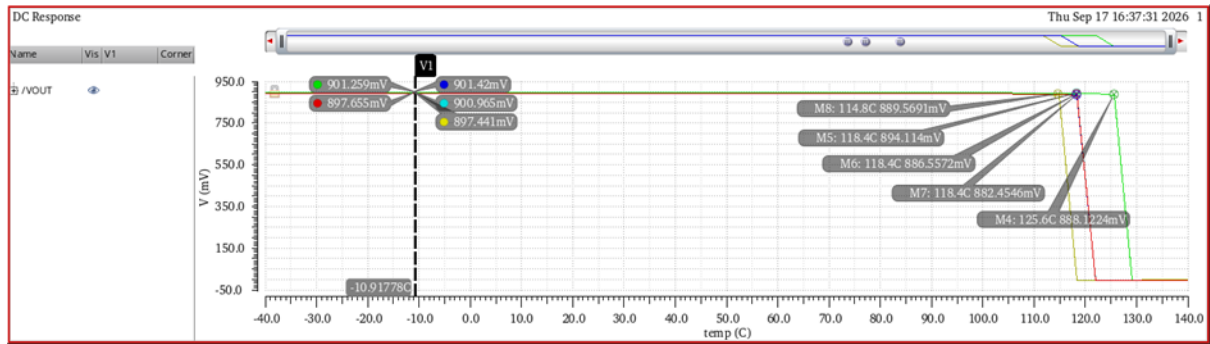


Power:

Name	Type	Details	Value
Iq	expr	$\text{abs}(\text{I}(\text{"/V2/PLUS"} \text{ ?result "dcOp"}))$	98.46u
Power	expr	$\text{abs}(\text{I}(\text{"/V2/PLUS"} \text{ ?result "dcOp"})) * \text{V}(\text{"/VDD...})$	118.2u

Thermal Limit:





Current Limit:

